

Representing data on a pie chart

The whole circle is the whole survey — every frequency becomes an angle.

LESSON 160–163

4 × 40 MIN

MATEMATIKA 6, §32

S7 16 INTERPRETING RESULTS

LESSON CLOCK

30'

35'

35'

35'

15'

10'

The whole is 360°	30 min
From frequency to angle	35 min
Drawing the chart	35 min
Percentages and pie charts	35 min
When the numbers are awkward	15 min
Homework	10 min

LEARNING OBJECTIVES

- Explain why the sectors of a pie chart must total 360° .
- Turn a frequency into a sector angle.
- Draw a pie chart accurately with a protractor and label it.
- Turn a percentage into an angle, and check the total.

TEXTBOOK

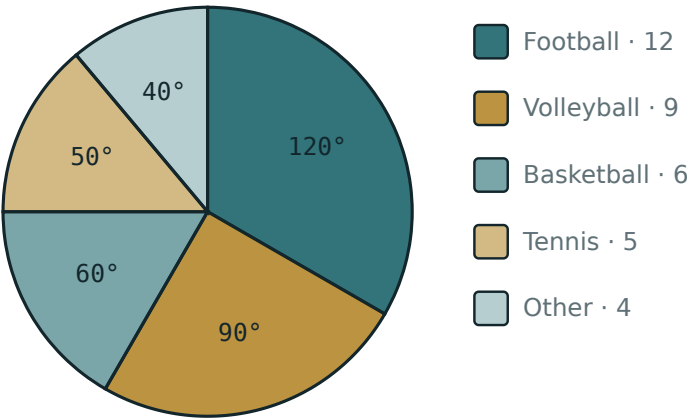
National	pp. 458–469
Cambridge	Stage 7, pp. 174–180
Workbook	Exercise 16.1

01

Explanation

The whole is 360°

A pie chart shows how a whole is divided. The whole circle — 360° — stands for everyone who was asked, and each sector takes the share of 360° that its group deserves.



$$\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360^\circ$$

FIND WHAT ONE PERSON IS WORTH, ONCE

With 36 pupils, $360 \div 36 = 10^\circ$ for each. Every angle is then the frequency times 10 — no fractions at all. With 60 pupils it is 6° each; with 45, 8° .

From frequency to angle

SPORT	PUPILS	WORKING	ANGLE
football	12	$12 \cdot 10$	120°
volleyball	9	$9 \cdot 10$	90°
basketball	6	$6 \cdot 10$	60°
tennis	5	$5 \cdot 10$	50°
other	4	$4 \cdot 10$	40°
total	36	—	360°

IF THE ANGLES DO NOT ADD TO 360°, SOMETHING IS WRONG

Add the last column before drawing anything. A total of 355° or 365° means an arithmetic slip, and it is far easier to find it in the table than in the drawing.

Drawing the chart

STEP	WHAT TO DO	WHY
1	draw a circle and one radius, straight up	a fixed starting line
2	measure the first angle from that radius	the protractor needs a base
3	measure each next angle from the line just drawn	the sectors join up
4	label every sector, or add a key	a chart without labels says nothing
5	write the total and the title	the reader needs the 36

DRAW THE LARGEST SECTOR FIRST

Small errors then accumulate into the smallest sector, where they matter least — and the last sector is what is left over, which is a free check on all the others.

Percentages and pie charts

A percentage is already a share of the whole, so it converts even faster: 100% is 360°, so 1% is 3.6°.

$\text{angle} = \text{percentage} \times 3.6^\circ$

PERCENTAGE	WORKING	ANGLE
50%	$50 \cdot 3.6$	180°
25%	$25 \cdot 3.6$	90°
40%	$40 \cdot 3.6$	144°
12.5%	$12.5 \cdot 3.6$	45°
10%	$10 \cdot 3.6$	36°

THREE LANDMARKS WORTH KNOWING BY HEART

A half is 180° , a quarter 90° , a tenth 36° . Any sector can then be judged by eye before the protractor confirms it.

When the numbers are awkward

Not every total divides 360 exactly. With 7 people each is worth $360 \div 7 = 51.4^\circ$ to one decimal place, and seven of those come to 359.8° .

TOTAL ASKED	ONE PERSON IS WORTH	EXACT?
36	10°	yes
45	8°	yes
50	7.2°	yes
7	51.4°	no — rounded
11	32.7°	no — rounded

ROUND THE ANGLES, THEN GIVE THE LEFTOVER TO THE LARGEST SECTOR

A pie chart must close. If the rounded angles come to 359.8° , add the missing 0.2° to the biggest sector, where it cannot be seen — and say in the working that you did.

02 Worked examples

EXAMPLE 1 Thirty-six pupils name a favourite sport: 12, 9, 6, 5, 4. Find the five sector angles.

① $360 \div 36 = 10^\circ$ for each pupil.

② Multiply each frequency by 10.

③ $120^\circ, 90^\circ, 60^\circ, 50^\circ, 40^\circ$.

They total 360° ✓

ANSWER

$120^\circ, 90^\circ, 60^\circ, 50^\circ$ and 40°

EXAMPLE 2 Of 60 pupils, 9 prefer tennis. Find that sector angle.

① $360 \div 60 = 6^\circ$ for each pupil.

② $9 \cdot 6$.

③ $= 54^\circ$.

Or $\frac{9}{60} \cdot 360$ directly.

ANSWER

54°

EXAMPLE 3 A survey gives 25%, 30%, 20%, 15% and 10%. Find the angles.

① 1% is 3.6° .

② $25 \cdot 3.6 = 90$, $30 \cdot 3.6 = 108$, $20 \cdot 3.6 = 72$.

③ $15 \cdot 3.6 = 54$ and $10 \cdot 3.6 = 36$.

Total 360° ✓

ANSWER

90° , 108° , 72° , 54° and 36°

03 Interactive model

Have the class find “what one person is worth” aloud before any angle is calculated; almost every mistake in this topic is a missed division at that first step.

Frequencies into angles

The sectors of a pie chart total:

100°

180°

360°

the frequency

With 36 pupils, one pupil is worth:

6°

10°

36°

12°

With 60 pupils, one pupil is worth:

6°

10°

60°

3.6°

12 out of 36 gives:

60°

90°

120°

12°

9 out of 60 gives:

45°

54°

60°

90°

25% of a pie chart is:

25°

90°

72°

180°

1% of a pie chart is:

1°

3.6°

36°

0.36°

If the rounded angles total 359.8° you should:

leave a gap

start again

give the extra to the largest sector

divide by 7

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Pie chart	Doiraviy diagramma	Круговая диаграмма
Sector	Sektor	Сектор
Sector angle	Sektor burchagi	Угол сектора
Frequency	Chastota	Частота
Total frequency	Umumiy chastota	Общая частота
Protractor	Transportir	Транспортир
Key (legend)	Izoh	Легенда
Proportion	Ulush	Доля

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angle of a sector is the frequency divided by the total, times:

With 45 people, one person is worth:

A sector of 180° represents:

A percentage becomes an angle when multiplied by:

3.6

36

360

1.8

Before drawing you should check that the angles:

are all equal

total 360°

are whole numbers

are under 90°

A pie chart without labels or a key:

is still complete

says nothing

is more accurate

needs no total

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The whole circle in degrees

ANSWER 360°

2. 36 pupils: the angle for one pupil

ANSWER 10°

3. 60 pupils: the angle for one pupil

ANSWER 6°

4. 12 out of 36

ANSWER 120°

5. 9 out of 36

ANSWER 90°

6. 6 out of 36

ANSWER 60°

7. The angles of a pie chart add to

ANSWER 360°

Medium · the standard question

7 PROBLEMS

1. 20 out of 60

ANSWER 120°

2. 15 out of 60

ANSWER 90°

3. 9 out of 60

ANSWER 54°

4. 6 out of 60

ANSWER 36°

5. 8 out of 24

ANSWER 120°

6. 3 out of 24

ANSWER 45°

7. 45 people: the angle for one

ANSWER 8°

Hard · stretch and reason

7 PROBLEMS

1. 25% of a pie chart

ANSWER 90°

2. 40%

ANSWER 144°

3. 12.5%

ANSWER 45°

4. 7 out of 40

ANSWER 63°

5. 13 out of 50

ANSWER 93.6°

6. 7 people in total: the angle for one, to 1 d.p.

ANSWER 51.4°

7. Why must the drawn angles be checked against 360° ?

ANSWER Rounding can leave the chart open

07 Homework

Homework — 5 tasks

Every pie chart needs a title, a total, and a label or key on every sector.

1. A survey has 30 pupils. Find the angle representing one pupil.
2. Draw the pie chart for frequencies 10, 8, 6, 4, 2 out of 30, with every angle labelled.
3. Of 200 people, 70 prefer tea. Find that sector angle.
4. Find the angles for the percentages 45%, 30%, 15% and 10%.
5. Draw the pie chart of your own survey data, with the total and a key.